

# Touch Device For Blind Reading

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**Abstract**—Vision is one of the main ways for humans to recognize information from the outside. Loss of vision or blindness hinders independence in life to a certain extent. Therefore, research on touch reading products for blind people is necessary. This study uses literature review, case analysis, observation methods, and qualitative and quantitative analysis. These three methods analyze the current development status of blind reading products in the existing market. In-depth understanding of the true reading habits of blind people through questionnaires. The research design uses audio and tactile reading to help blind people input and output information through the above analysis. Combining the three input methods of computer, mobile phone and memory card to help users obtain information through braille display and voice playback. Use voice assistants and braille keyboards to help users output information to meet their life and learning needs. The product has a two-layer design, namely the input layer and the output layer, which can expand the braille display interface and improve the user experience. Finally, it is presented in the form of product renderings and product scene diagrams.

**Keywords**- Blind, Assistive Design, Braille, Touch Reading, Electromagnetic Braille Display, Sensory ompensation

## I. INTRODUCTION

The improvement of quality of life owes develops technologically, and spiritual pursuit becomes a part of life. Reading has become a vital link in people's lives. We receive information every day in different ways, such as mobile phones, computers, iPad, newspapers, and magazines. the importance of reading is obvious.

As of 2017, more than 2 million people in the UK have lost their sight, including 1.7 million people with visual impairments in England. According to statistics, 4.1 million people will be blind by 2050 and affect their normal lives. Take London as an example. 208,000 people have visual impairments in London. In the working-age group, only a quarter of blind and low-vision people have employment opportunities [7].

According to RNIB (Royal National Institute of Blind People) statistics, "2 million vision loss is caused by 39% of refractive errors, 23% of age-related macular degeneration, 19% of cataracts, 7% of glaucoma, and 7% of other eye diseases. 5% of diabetic retinopathy. "Reflex error is the leading cause of the most critical vision loss [8]. Blind and visually impaired people have great employment problems because of the missing parts of their bodies. There are many reasons for this problem, such as low education level, need professional help at work, and low literacy skills. These reasons reduce the probability of working for blind people. Some people with eyesight get less paid work than healthy people. These have caused them to be

ignored in society. The way to improve these phenomena is to improve their office skills, use auxiliary tools to complete their work independently, and learn Braille. These measures can not only improve the self-confidence of the blind and visually impaired, but also improve the quality of life of the blind, improve the efficiency of work and study, and the ability of the blind to independently obtain information. As a special group of blind people, they need a particular way to read books. Braille publications are paper publications with raised symbols for the blind to touch with their fingers. On the other hand, Braille information services have become a topic of concern for designers and researchers in modern society. Reading aids for blind people with different functions have appeared, for example, a braille display.

Although the hearing is also the main way to obtain information, touch reading and braille can effectively help them study and work independently. This research will focus on designing touch-reading products for blind people, so that blind users can read more comfortably and conveniently, and meet the needs and reading habits of blind people. This research deeply understands blind people from different perspectives, such as psychological, physical, behavioural and social characteristics. Provide a favourable basis for the design process. The Blind Touch Reader is a product that helps the blind to read information. It is not only a product but also the best assistant for the blind when reading, such as guide sticks and guide dogs. They are all tools to help blind people solve daily life.

Therefore, the braille touch reader has crucial social value. The design is mainly based on the development of existing touch-blind readers from the perspective of the physiological, psychological and social characteristics of blind users. Consider the main functions of the product from the perspective of practicability and convenience. Design a user-centred reading device to improve user experience and help blind people to learn daily through touch and hearing. This can not only improve the reading resources and reading experience of blind people, but also improve the quality of life of blind users, increase employment opportunities, and satisfy the ability of blind people to obtain information actively.

## II. RESEARCH METHODOLOGY

This research mainly uses the "double diamond method". The method is mainly divided into four stages: discovery, definition, development, and delivery. In the discovery stage, learn about the development of touch readers for blind people, the technical support of touch reading devices for blind people, the definition and influence of sensory compensation, the development and current situation of braille, the psychology of

blind people. The method of literature review mainly collects and sorts out relevant literature, obtains relevant information on this topic and understands the research status. In the definition stage, the case analysis of blind touch reading products in the existing market, the design ideas of the product cases are integrated into this research. The case analysis method understands the use of the product itself and users by blind readers and conducts a comprehensive analysis of the current market for blind readers. Summarizing the pain points of existing market products is conducive to the determination of design schemes and design ideas. In the development stage, some user opinions are collected through questionnaires and in-depth interviews, and the functions and details of the product are discussed with users in depth. The observation method can understand the needs of users and deeply experience the real life of the blind. Observe the physiological and social characteristics of blind people. Focus on the parts that need to be optimized in the use of products for the blind. The final stage integrates all the information and proposes some design concepts to determine the final design plan. As show in figure 1.

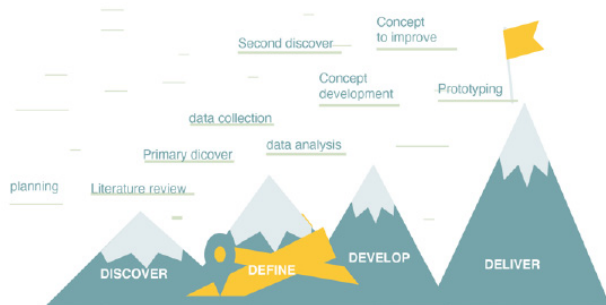


Figure 1. the double diamond method plan

#### A. Literature Review (Secondary Research)

Sensory compensation refers to the phenomenon that when some of the human senses are damaged, other sensory functions will be enhanced accordingly. For example, blind people are more sensitive to hearing and touch than ordinary people. People's knowledge of things begins with feelings. Human senses are sight, hearing, touch, smell, and taste. To understand external things, people need to organize sensory information and form an overall response to the real thing. Confirm what it is through human brain cognition and incorporate it into the conceptual category. This is perception. (Carterette and Friedman, 2014). Sensory compensation comes from a series of cognitive processes such as human perception, perception, imagination and memory, and their interaction with each other.

The process by which people acquire knowledge or information processing is called cognition. This is the most fundamental psychological process of people. It includes feeling, perception, memory, imagination, thinking and language. The human brain accepts information input from the outside world, processes it, and converts it into mental activities and dominates human behaviour. This is the cognitive process. Therefore, people with visual impairments have stronger hearing and touch functions than ordinary people. They can

remember people or things through a sound and can acquire knowledge using touch braille.

The educational deficit of the visually impaired is not a lack of visual experience. The blind need to learn how to use touch to move, read and communicate. The consequences of early and late blindness have an impact on rehabilitation and education. However, people with visual impairment use the sense of touch to act, and it takes longer to read with the touch than with the eyes. Vision is a crucial sensation and provides a lot of sensory and perceptual stimulation. Visual and intuition deprivation has a devastating effect on a person's mental health. The lack of visual input can change people's response to touch and may cause great psychological distress. For example, people who have just gone blind have sleep disturbances during the process of adjusting to the dark. (Heller and Gentaz, 2014)

The function of the blind reader is to transform the content of printed materials into auditory or tactile forms perceivable by the blind. The first photoelectric device for blind people was born in 1912. In the 1960s, text and image recognition technology gradually matured. In 1966, Hans A. Mauch and Glendon C. Smith of Mauch Labs collaborated to invent the Visotactor, a blind reader with touch mode output. The user perceives the number of vibrations and the vibration time to obtain letter information through fingers. During the same period, Optophone and Lexiphone marketed blind readers with auditory output. However, users need professional training to master these codes. In order to provide users with a more convenient experience, Mauch Lab designed the spelling voice reader Cognodictor. Then came early speech synthesis. With the development of text recognition technology and TTS technology, blind readers have made breakthroughs in technology. In 1975, Kurzweil invented a KRM (Kurzweil Reading Machine), which can read the information users need. Difficult to use due to the size and weight of the machine. The appearance of the blind reader has changed with the development of technology. The core of the development of blind readers is the needs and behaviours of users. The bulky reading devices have evolved into handheld devices. The demand for portable products has increasingly become the development trend of reading products.

The main function of most of the current products is to convert the text in the book into speech, and use hearing to get the content of reading. It is mainly to use text recognition technology to collect the content of reading, and then extract the text through image processing technology to recognize the text and output the sound. So, the main technology is to be able to convert text to speech, text to Braille, and recognize text. Text recognition technology (OCR) for blind readers. The research and development of tactile perception products mainly focus on tactile stimulation, which is a specialized application research of tactile reproduction technology in virtual reality. The tactile display terminal mainly includes electric stimulation type, electric heating type and electromagnetic type.

#### B. Case Study (Secondary Research)

Wimmer and Dominick (2000) pointed out that when researchers want to know much information about the project, case analysis is the most valuable method. The case analysis method focuses on analyzing products in the current market and

understanding the technologies and functions already available in existing products. Get blind people needs to optimize the use of reading products. This study chose two cases to compare and contrast the blind readers in the market. The product examples have chosen BraillePen 12 Touch and VarioUltra 20/40.

### C. Observation method (Secondary Research)

Observation method is one of the critical links in the design process. This method can deeply understand the daily life of the blind and understand the physiological and social characteristics of the blind. At the same time, from a product point of view, we need to understand the main ways blind people obtain information. We can master the parts that need to be optimized in the use of reading products, and design and improve user experience. This study observed a young lady (Anmy) who was blinded by accident. She did not give up life because she needed to take care of her family. She uses some auxiliary products to complete daily life, such as guide sticks, smartwatches, braille readers, and special smartphones.

### D. Qualitative methods (Primary Research)

#### 1) in-depth interview

In-depth interviews can have irregular questions and timing, which is helpful for researchers to understand the real-life environment and product needs of users deeply. Researchers can guide interview trends based on the basic questions required by the project. The interviewer can expand the additional information they want to tell the researcher based on the situation of the question. Interviews may be understood as a form of chat so that researchers can collect the real thoughts of users.

### E. Quantitative methods (Primary Research)

The total number of questionnaires collected in this study is 194, and the questionnaire lists some of the central answers to facilitate user choices. The questionnaire was designed for customers with 23 questions to investigate users' habits and needs, as well as users' perceptions of blind people reading products. Send the questionnaire to the user or the user's family by e-mail. To deeply understand the user's habits and feelings when using the product, the user's basic information. Due to the particularity of the survey population, this study combines open and closed answers to understand blind people. The questionnaire is composed of background questions, subjective questions and test questions. As show in figure 2.

| Basic Information             |                        |             |         |
|-------------------------------|------------------------|-------------|---------|
| Statistical Indicators        | classifying indicators | Sample Size | Ratio/% |
| Age                           | ≤18                    | 19          | 9.7%    |
|                               | 18-34                  | 43          | 22.1%   |
|                               | 35-55                  | 102         | 52.5%   |
|                               | ≤56                    | 30          | 15.4%   |
| Income                        | High Level             | 31          | 15.9%   |
|                               | Medium Level           | 96          | 49.4%   |
|                               | Low level              | 67          | 34.5%   |
| Duration of blindness (Years) | ≤1                     | 12          | 6.1%    |
|                               | 1-5                    | 19          | 9.7%    |
|                               | 5-10                   | 70          | 3.6%    |
|                               | 10+                    | 93          | 47.9%   |

Figure 2. Basic Information of customers

## III. PRINCIPLE FINDINGS

### A. Literature review

Design is a complex project. Carrying out the project in an appropriate and timely manner is a prerequisite for developing successful design results. The phase is to plan and prepare to do the right thing before the researcher starts to do the right thing. It also sometimes involves rethinking what is done in the project when an accident happens. (VAN BOEIJEN, 2020) Therefore, in the literature review part, we mainly understand the development and technical field of the blind touch reader. The influence and effect of sensory compensation on people with physical disabilities. Psychological changes of the blind in the early and late stages and a guide to the use of braille. Use this information to accumulate design thinking, find a breakthrough in design and understand the needs of blind users. This also helps to understand the design concepts and design functions of blind readers in the market.

### B. Case study

- The Braille pen12 touch



Figure 3. BraillePen 12 Touch

Braille pen12 touch can be connected to a smartphone, computer or other devices to obtain information. (PC, Mac, iOS, Android, Symbian;) It can browse various information on external devices by inputting and outputting Braille. The weight of this machine is 260 pounds, and the size is 150 x 94 x 25 mm, which meets the needs of users to carry around. This machine is Supported by Voiceover, talks&zoom, mobile speak, Mobile Accessibility, brailleback, Window-Eyes, Supernova, NVDA, System Access, ergonomic keyboard. Use Bluetooth technology to connect to the user's mobile phone, PC or iPad for browsing external device. Use the six-key keyboard to write text messages, browse the Internet or read browsing information. Switch between 6-spot braille and 8-spot braille through the braille display. In summary, it meets the basic needs of the blind. However, the small braille display increases the user's reading time and reduces the user's reading efficiency.

- VarioUltra 20/40



Figure 4. VarioUltra 20/40

VarioUltra has two models of blind readers (40&20, its function focuses on word processing. The advantage of this product is that it can be connected to four Bluetooth devices and a USB device at the same time. Noteaker equipped with a secondary translator can open DOC, DOCX, RTF, TXT, BRF, BRL, XLS, XLSX, PDF file spreadsheet viewer and PDF viewer calculator, scientific calculator and file manager with ZIP archive time check, date indicator, alarm clock, stopwatch and timer. There are six display buttons in total, three on each side. These buttons are used to browse documents while the user is reading. These buttons are also used to command and control certain devices, such as iPhone or iPad. These two blind readers use 8- button braille input keyboard with two ergonomic space bars. The weight of VarioUltra 20 is 320 grams, and the size is 18.8\*8.9\*2.0cm. The weight of VarioUltra 40 is 530 grams, and the size is 31.6\*8.9\*2.0cm. Both products also have 32GB flash.

The two products are suitable for most users and meet their basic needs. In terms of functions, the touch-reading products for the blind can be connected to mobile phones and computers via Bluetooth to help users obtain information from computers or other devices. Use input and output braille to browse various information on external devices. Whether it is reading or writing, it can help blind users complete the operation. For the blind, hearing and touch are the most important ways to obtain information. However, the two products focus on braille displays and how to use braille to obtain information, ignoring the way of hearing information. How to use the product to meet the user's usage habits and usage methods, on this basis, to reduce the user's pressure is a major issue. The focus of this research is to design a user-centric reading device that improves user experience and helps blind people to conduct daily learning through touch and hearing.

### C. Observation Surveys

The purpose of the observation method used in this study is to understand the real life and living habits of blind people. Most blind people are blind for some reason in modern society. They adapt to the new environment from the light to the dark life in front of them. In this process, they have to familiarize themselves with their living environment. Try to handle some things independently in this environment. The person observed in this study was an assistant before she went blind and had a complete family. Because of unexpected changes in her life, what she needs to do is to adapt to life and use some auxiliary products to help her live independently. It took her a year to

slowly accept the facts and find a job that she can do, which is to help her newly blind companion adapt to her new life in the blind community. The following pictures are her life storyboard.



Figure 5. the storyboard 1

In the morning, she needs to do housework and use iWatch to know the weather and call, because Siri and Voice over are both perfect life assistants. After the family went out, she began to prepare the essentials for travel. She had to find a walking stick and smart glasses, and then bring a braille reader. As show in figure 5.



Figure 6. the storyboard 2

She enjoys breakfast in a restaurant that she frequents. In the restaurant, she will hear some interesting topics. As show in figure 6.



Figure 7. the storyboard 3

In the community, she and her friends did some simple dance moves. She believes this can keep her body healthy. As show in figure 7.



Figure 8. the storyboard 4

She taught her companion to learn braille in two hours of reading time. She knew braille very well because she lost her eyesight ten years ago and was used to dark times. She used braille to get the information she needed. Although the price of smart braille reading products is high, it is a necessity in her life. She will use the machine carefully to extend its service life. As show in figure 8.



Figure 9. the storyboard 5

Volunteers will help them go for a walk outdoors after the community activities, and go to the garden with friends to enjoy the sunshine. As show in figure 9.



Figure 10. the storyboard 6

Anmy finally went home. Although she lost her eyesight, it did not affect her life. " As show in figure 10.

A series of mental activities usually accompany daily physical exercise, and blind people are no exception. The

physical disability and the surrounding living environment cultivate the blind optimism, positive spirit and fear of the future. Of course, they will also have a more vulnerable side, mainly due to congenital physical defects and characteristics that lead to this unique behavioral characteristic. Ordinary people may feel anxious and fearful in the dark, which seriously interferes with the normal life they want to live long-term. Their physical disabilities make them feel unsafe because it takes a long time for them to adapt to the changing environment. They need some time to adjust.

In daily life, if blind people are in an unfamiliar environment, they usually feel helpless and at a loss. After observation, it is not that blind people cannot take care of themselves as we think. After long-term training and habits, they can live normally in their environment. A blind person can live a normal life when moving in a familiar space after observing the daily life of the blind person. For example, blind people can do simple things in the kitchen like ordinary people. Blind people can use mobile phones; they can also feel where they are and find things they often use.

The most important part of a blind person's life is socializing and entertainment. These two parts enable them to get the care and help from family and friends. Their other feelings are more subtle and obvious after losing their eyesight. Therefore, when they obtain information through the Internet, they need a convenient and accurate touch reading product.

#### D. Questionnaire Surveys

Types of information obtained by the blind

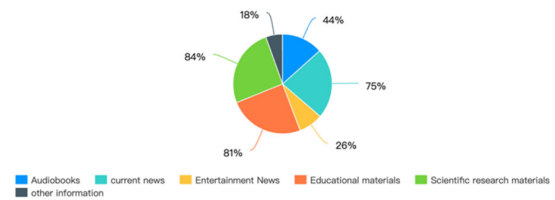


Figure 11. types of information obtained by the blind

Ways for the blind to obtain information

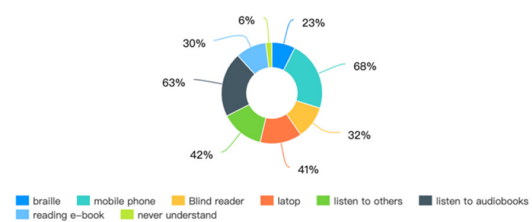


Figure 12. ways for the blind to obtain information

Audiobooks have replaced Mp3 players along with the development of technology that can be connected to mobile applications. For example, literary books, philosophy books, and novels can all be accepted through mobile apps. The Internet has replaced separate audiobook playback devices. Audiobooks are useful learning tools for the blind, and there



will be no great pressure in the listening process. Sound can convey thoughts and emotions.

The data shows that their reading style is no different from health people for most blind people. They are very willing to know current affairs news and entertainment information. The touch of the blind is more profound than that of ordinary people. Ordinary people can intuitively understand the essential characteristics of things by observing things with the naked eye. Such as appearance, shape. Blind people can perceive things through touch.

On the other hand, hearing is also one of the crucial ways for blind people to perceive things. In the dark, the voice resolution of a blind person may be higher than that of an average person. They extract what they want from the noisy sound. For example, blind people can recognize the source and direction of sound through hearing. Therefore, hearing and touch are equally crucial for blind people.

Blind people can use Braille to read to enhance their memory and deepen their understanding. Blind people can exercise their ability to use language through learning, such as expressive ability and creative ability. Braille enriched their spiritual needs. In other words, Braille can enable blind people to learn and work independently. They can get more job opportunities.

#### E. In-deep Interview



Figure 13. respondents (Yalan Liu)

The subject of the in-depth interview in this article is a blind masseur who has certain cognitive abilities. She is full of hope in life and focuses on her professional field. The content of the interview is mainly basic information, the duration of blindness, the main ways and means of obtaining information, the level of Braille learning, and the need for reading content. This research uses an in-depth interview method, and the interview content is flexible. This allows researchers to learn more about interview trends. Blind people are different from the visually impaired. Their eyes are completely dark, so they form a unique way of perception and expression. As members of society, they have the right to integrate into society and be respected. Therefore, understand their daily behaviour and the way of using reading products, as well as the problems that arise during use, and analyze the actual needs of blind people and expectations of reading products. These are the key parts of designing a touch reader for the blind. Products can accurately meet real needs through communication and truly solve their life problems. In

the interview, they found that they are friendly to strangers and are very willing to communicate about things in life. They do not want to live as we think they cannot take care of themselves. They can usually live in their living environment after adapting for a long time. For the blind, they also cannot live without electronics. They communicate with other people through the Internet. They can also learn and shop online through the Internet.

## IV. DESIGN AND DEVELOPMENT

### A. Design features

This research mainly focuses on four aspects: user needs, pain points when using the product, product functions, and appearance layout. This study understands the needs of blind users and the blind touch reader through literature review, observation method, qualitative and quantitative analysis, and interview method

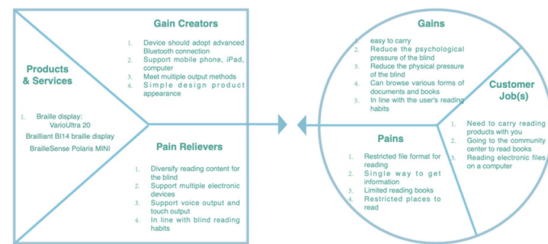


Figure 14. the value proposition canvas

#### • User needs

This research focuses on the diversification of blind reading content, the comfort of reading methods and reading habits, and the basic needs of users for products through preliminary research. Appearance comfort and portability are the core of the product. A simple layout is considered in the interface design of the reader because the complicated interface design will increase the burden on the user and bring confusion to the user. The purpose of having a simple and easy-to-use functional design and a simple interface is to have a comfortable user experience. Users can understand and remember and develop natural usage habits. The first step is to design according to the physiological characteristics of the user group, and the braille keyboard should conform to ergonomics. Have an in-depth understanding of their needs through primary and second research. The appearance brings memory or operational burden to the target user, which can improve work efficiency, safety and comfort.

#### • Pain points when using the product

The rationality of product functions can make users feel relaxed when using the product. This not only facilitates their lives but also facilitates reading. Ensure that the actual user requires the function; there are multiple additional functions, consider the necessity and rationality of other functions. Make the product suitable for users of different ages and meet the usage habits and product expectations of blind users. Use reasonable interaction methods to enhance communication between users and products. Therefore, in the process of designing a product, we must not only pay attention to the

product but also understand the behavior of people when interacting with the product. It is achieved through certain technical means, such as voice recognition, text recognition technology, voice search and other multimedia technologies. Broaden the application scope of barrier-free design for blind people, provide a more convenient lifestyle for blind users, and realize barrier-free communication between people and products.

#### B. Idear generation

This research mainly focuses on four aspects: user needs, pain points when using the product, product functions, and appearance layout. This study understands the needs of blind users and the blind touch reader through literature review, observation method, qualitative and quantitative analysis, and interview method.

- Concept 1.

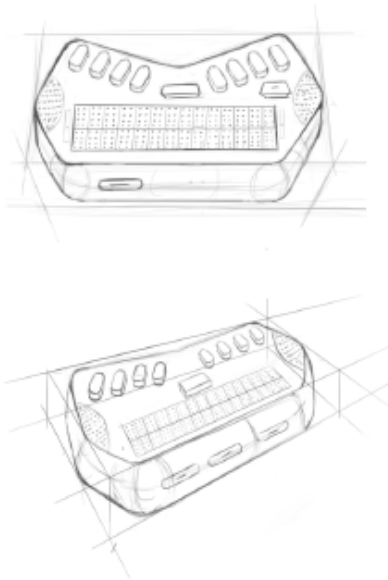


Figure 15. . Concept sketch 1

The first concept uses hearing and touch to obtain information. The design concept reduces the complexity and interface of the device. Improve the efficiency of touch reading by adding Braille cells. The unique shape conforms to the posture of the hand, thereby improving the ability to recognize the blind. In the normal operation of the product, it can conform to the user's mental model.

- Concept 2.

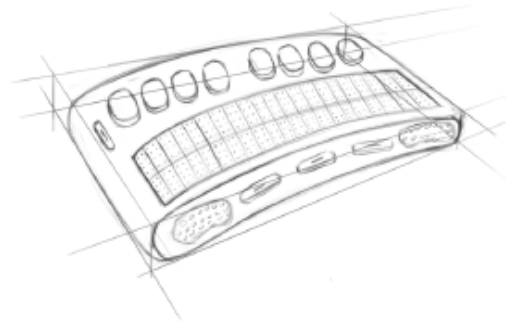


Figure 16. Concept sketch 2

The second concept mainly removes those function keys that may interfere with the normal operation of the blind and makes the function of the product easier. It conforms to the cognitive psychology and usage habits of blind people, and provides a functional layout. Reasonable interaction can bring better user experience. In the early stages of design, we focused on the needs of users and the behavior and habits of blind users when interacting with the product through appearance layout. The extra button is designed on the side of the product.

- Concept 3.

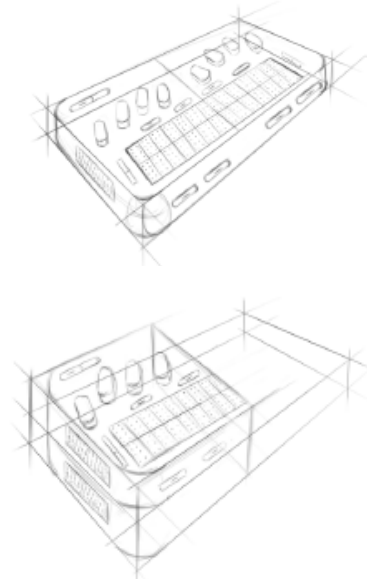


Figure 17. Concept sketch 3

The third concept is to add some functions that may be needed based on basic functions, for example, recording function, radio. The concept uses a folded form to achieve portability. Added broadcasting function and recording function. Audio storage extends time continuity for users who need to work. In order to improve the convenience of reading the product, the product is designed in a folded form.

- Concept4.

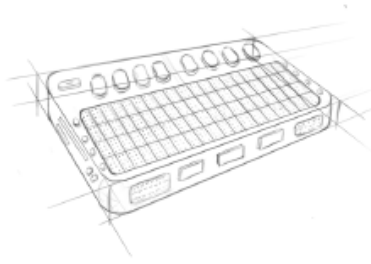


Figure 18. Concept sketch 4

This concept is mainly to increase the braille unit to maximize the efficiency of blind reading. For the convenience of users, a local storage function has been added. Data from other devices can be transferred to the device via Bluetooth, and files in various formats can be read, such as PDF, TXT. Three computer keys Shift, CTRL and ALT have been added to facilitate the synchronous use of the product and the computer.

### C. Concept evaluation

In-depth study of the feasibility of the idea selected the most suitable idea. Because there are many different concepts, a multi-standard methodology was chosen to better support and improve future actions. The selection criteria are determined in this method. Consider the importance and priority of these concepts, and assign weights to each criterion (up to 100%). These standards are selected according to the goals and objectives described in the introduction chapter. During the research phase, it is not difficult to find that most blind users have two perceptual reading habits: hearing and touch. With the rapid development of modern technology, blind users have become accustomed to the general trend of society. Their reading methods mainly include mobile phones, radios, and readers. In the survey results, more than 75% of these people are more willing to read professional-related information, news and some entertainment news. Therefore, the reading content of blind people is the same as that of ordinary people. Most blind users have deep reading desires and needs. They prefer to obtain information through mobile audiobook apps and radio. However, when they need more professional information, these devices cannot satisfy them. For blind users, the initiative of mobile phones to obtain information is not strong enough.

It is necessary to evaluate the appearance of a conceptual product during the product design process. After having some preliminary design ideas, combine research and theory to evaluate the concept, then select and combine the most suitable design plan and develop it. Integrate sketch plans through comparison and determine product standards. Then consider the importance and practicality of each sketch. The suitability, function, size, appearance and difficulty of the prototype were selected. The selection and positioning of functions can bring different experiences to users. The size is also an essential part of the product design process, and the size of the product must conform to the user's habits. The appearance is more concerned about the safety of the product. At the beginning of this chapter, it was mentioned that blind people are easily injured because they lose their eyesight. Therefore, the design is worthy of further evaluation and selection. The percentage distribution is

as follow: suitability: 20%, functionality: 30%, size: 20%, appearance: 20%, difficulty of prototype production: 10%

|           | Suitability | Functionality | Size | Appearance | Difficulty of prototype production | Total score |
|-----------|-------------|---------------|------|------------|------------------------------------|-------------|
| Concept 1 | 3           | 3             | 4    | 4          | 5                                  | 3.4         |
| Concept 2 | 4           | 3             | 4    | 3          | 5                                  | 4.1         |
| Concept 3 | 3           | 4             | 5    | 4          | 4                                  | 4.2         |
| Concept 4 | 4           | 5             | 4    | 4          | 3                                  | 4.3         |

The first concept scored the lowest among the four concepts. From a practical point of view, it only has the most basic purpose. There are two forms of Braille output and input. The shape of the product looks more space-saving, using the appearance to make a posture that matches the hand. The generation of this model is relatively simple, and it has insufficient functions for blind users.

Concepts 2, 3, and 4 are high-scoring design concepts, which are developed through two key variables of appearance and function. The function of the product is most important in two variables. The second concept considers the comfort of the hand when carrying it around. Three side buttons have been added to transfer the three functions of the computer buttons to the blind reader, which is convenient for the product to connect to the computer and add voice functions. From the perspective of user experience, since the product is presented in a curved form, it may be uncomfortable to use.

The third concept comes in two forms. This foldable form allows blind people to touch and read the product in a smaller size. Its advantage is that it is easy to carry, but there is no substantial change in function. Model making is relatively tricky.

In terms of functions, the fourth model adds a memory card reading function, a radio, a button for adjusting the volume, a speaker and a button for synchronizing the computer. The increase of braille units has improved the efficiency of reading for the blind.

Through comparison, Concept 1 and Concept 4 are combined together, and multiple applicable forms of Concept 3 are adopted. The elements of these three concepts are combined to form a multifunctional blind reading device.



#### D. Idea development

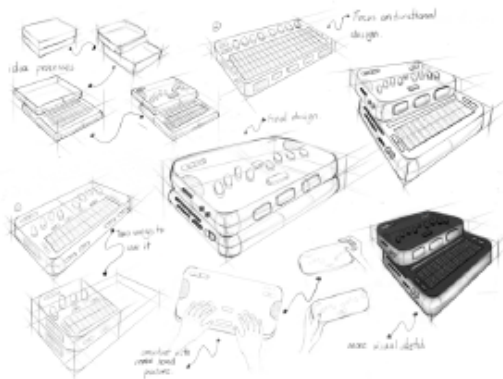


Figure 19. idea development sketch 1

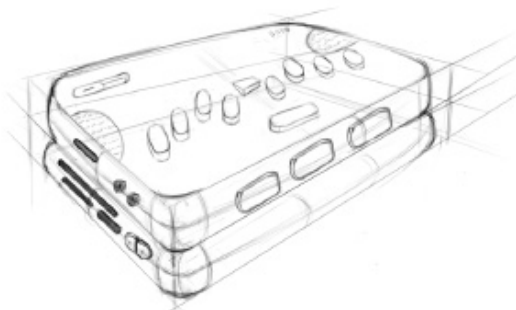


Figure 20. idea development sketch 1

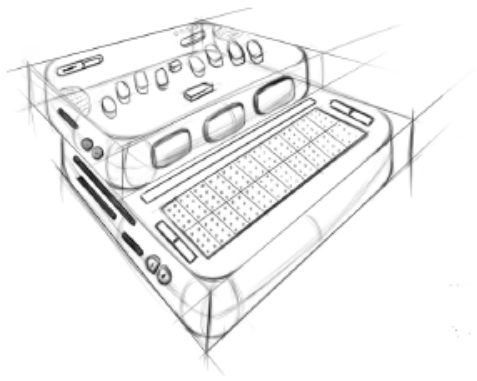


Figure 21. idea development sketch 2

The idea is a device that combines touch and hearing. The device can input and output in two ways, increasing the conditions for actively obtaining information. Users can use memory cards, Bluetooth to receive audio and files from other devices. Use Bluetooth or WIFI to connect to mobile phones, PC, Mac, iPad. The input mode is braille keyboard and voice search and supports reading computer files such as DOC, DOCX, RTF, TXT, BRF, BRL, XLS, XLSX, PDF file. Store or obtain information and convert files in two ways to meet the needs of blind people who study and work every day.

Search for information in three forms, voice search, braille input, and photo search. Users can use three search methods to obtain information. Users can choose the way of using the information according to the surrounding environment. Voice search is no stranger to the current society. Siri was founded in 2007 and was acquired by Apple in 2010. At first, it was mainly based on text chat services, and then through cooperation with Nuance, the world's largest voice recognition company, Siri realized the voice recognition function. Now Siri can not only help users open simple files through voice, but also directly search for information. So, adding a voice assistant to help users search and obtain information is one of the essential functions of this product. As show in figure 22.

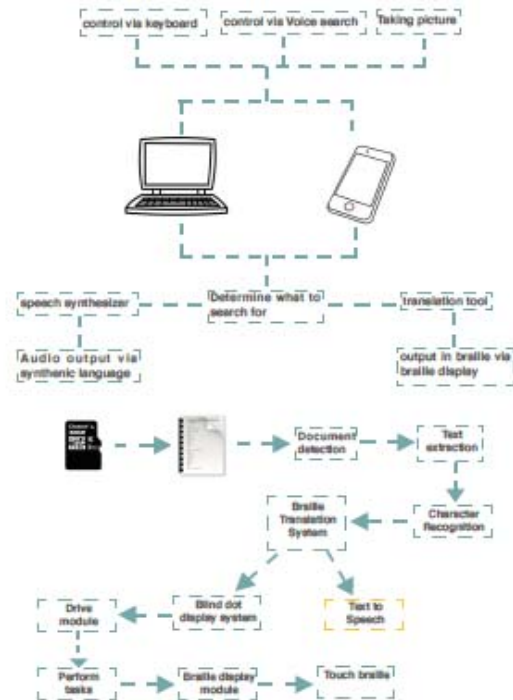


Figure 22. . technical structure design of operating device

The form is to use the camera to lock all the contents of the current page. In addition, the information is preprocessed with text and images. The images captured by the camera will be a little distorted, skewed, and some spam messages have nothing to do with the text. In order to ensure that the image quality meets the requirements of text recognition, improve system recognition efficiency and better recognition effect, it is necessary to preprocess the input image. Preprocessing mainly includes grayscale, smoothing and denoising, binarization, word segmentation, normalization and refinement of the original color image. After confirming the information, the device converts the content into braille or uses voice playback to obtain the content.

On the other hand, the memory card obtains information and inserts the memory card into the card slot to detect the required files. The machine scans all the contents of the document. Finally, the text image preprocessing function extracts the text and outputs the text in speech and braille respectively.

### E. Design function

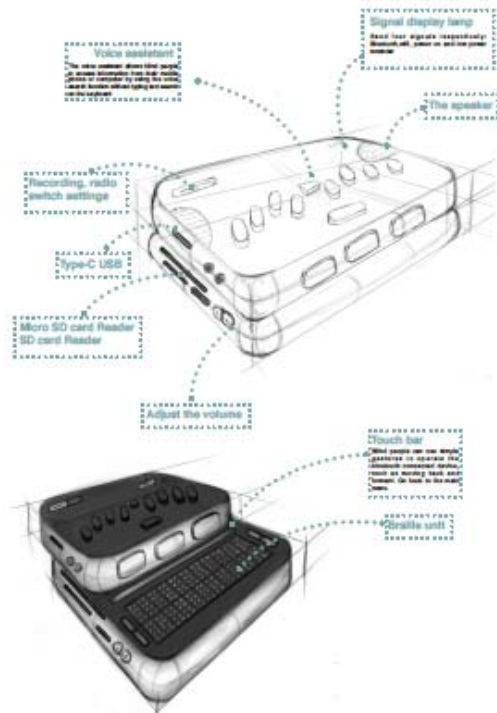


Figure 23. Product function introduction

This idea consists of three basic concepts, with two forms of input and output, namely hearing and touch. On this basis, a voice search assistant, recording function, radio and speakers are added. In addition, an embedded input method, SD card, and USB file reading functions have been added. Users do not have to worry about the restrictions of the location and environment when they obtain files. The purpose of adding SHIFT, CTRL, and ALT buttons is to adapt to computer equipment. Expand the scope of barrier-free design and application for the blind, provide a more convenient lifestyle for the blind, and realize barrier-free communication between people and products to make product use more reasonable.

- Voice search assistant

Language has strong descriptiveness as a way for people to communicate, which is more convenient than keyboard input. The user needs to enter characters and then manually click the search button. Voice services are characterized by high efficiency, convenience and are suitable for users who are not familiar with Braille.

- recording function

The function of recording is mainly to help blind users collect information and save data in time. Blind users can use the recording function to collect important information.

- Radio

Listening to the radio can relax the user's body and mind, receive local radio stations and learn about current hot topics. It is one of the tools to integrate into the social environment

because the life circle of blind users is smaller than that of ordinary people.

- SD card to get information

SD cards are widely used in current consumer digital devices. SD card is a kind of multi-function memory card, which has the characteristics of large capacity, high performance and safety. Furthermore, easy to reformat, so it has a wide range of applications. Multimedia files (such as music, movies, news) can be easily saved to the SD card.

- Touch bar

The touch bar helps the user to continue to the next message using swipe gestures. Blind people can use simple gestures to operate Bluetooth-connected devices, such as moving back and forth. Return to the main menu.

The internal structure of the blind reader is mainly divided into five parts: embedded system, TTS speech synthesis, text recognition technology, voice search, electromagnetic blind spot display technology, and touch bar. The embedded system is a small microprocessor whose primary function is to process some data. Embedded system consists of two parts: software and hardware. The hardware mainly includes the processor, memory and connecting external devices. Speech synthesis technology calculates and processes some digital signals, acoustics and language, and uses voice output to enable blind users to receive information in different environments. Text recognition is mainly to preprocess images and extract text. Voice search technology reduces conventional search steps, improves search speed, and can quickly perform voice search without entering text.

### F. Selection of materials

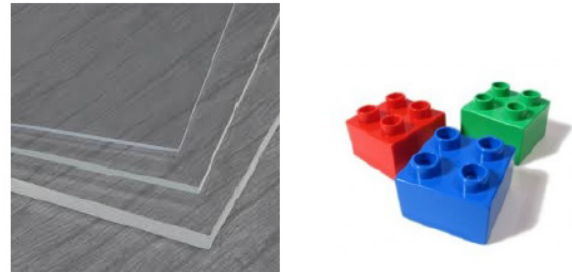


Figure 24. PMMA & ABS

When selecting materials for the device fully consider the lightness and durability of the materials. The most common one is plastic. There are many types of material, such as ABS and PC. In order to achieve product aesthetics, the cost and technical process of the material itself should be considered in the design of blind readers. Polymethyl methacrylate (PMMA) is a critical plastic polymer material developed earlier and has good chemical stability. And transparency. It is also easy to dye and process and is also widely used in the construction industry. (Acrylonitrile butadiene styrene) ABS resin has good heat resistance, impact resistance, chemical resistance, low-temperature resistance and good electrical properties. The advantages of surface gloss and product size stability can be used to color the material surface, such as spraying metal on the

surface, electroplating, welding, hot pressing and bonding. It has been widely used in textile, electronics and other fields.

*Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

#### G. Braille point display driver selection

The electromagnetic braille display device provides support and requires a lower voltage. electromagnetic blind spot display device mainly uses the electromagnetic coil's magnetic field when the power is turned on. The magnetic field disappears after the power is turned off. Then through a specific mechanical structure, the braille contacts are raised and lowered to form braille characters. The braille display devices on both sides are composed of eight dots, and the corresponding driving mechanism is arranged in three layers. electromagnetic braille display device has a fast refresh rate, high reliability, low price, and convenient manufacturing and maintenance.

#### H. Solidworks Modeling display



Figure 25. Solidworks modelling display 1

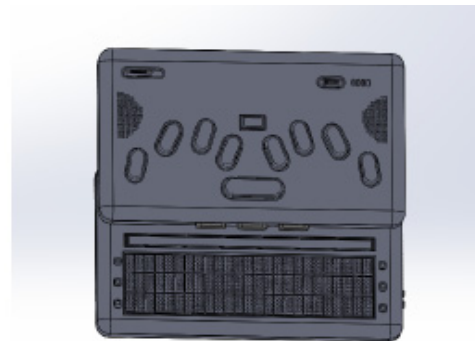


Figure 26. Solidworks modelling display 2



Figure 27. Solidworks modelling display 3

#### I. Product size design

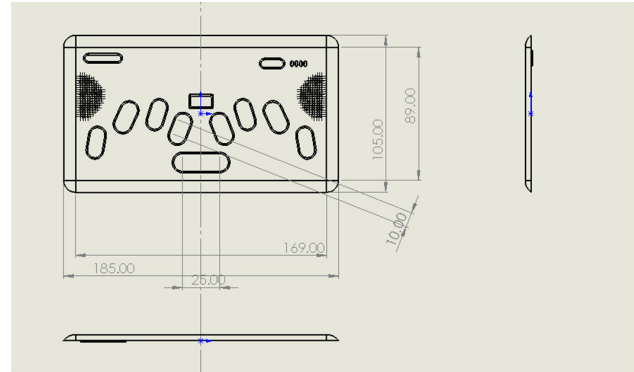


Figure 28. product design 1

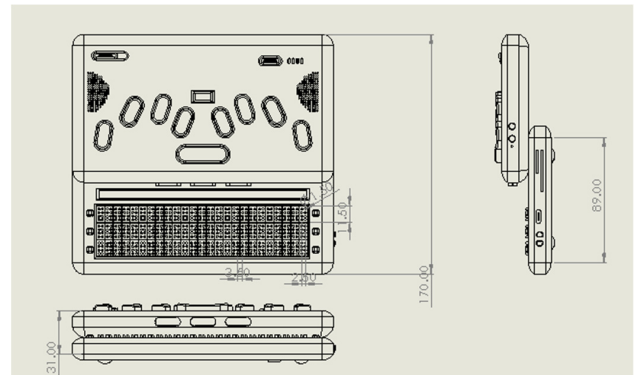


Figure 29. product design 2

The size is suitable for the holding posture of the hand and improves the portability of the product. The overlapping size of the product is 185\*105\*31mm, and the expanded size is 185\*170\*31mm. In order to make the input and touch feeling of blind users different, the size of other buttons is different from the size of the braille keyboard to distinguish the buttons and improve the input speed of blind users. As show in figure 28&29.

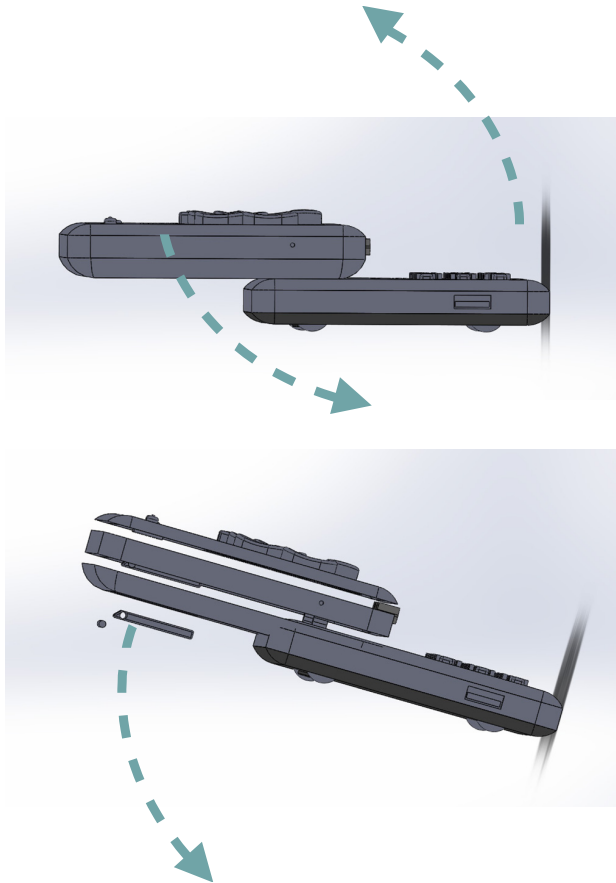


Figure 30 & 31. design solution model by SolidWorks

Since the first layer has more parts than the second layer, it weighs more than the second layer. When the user uses the double layer, the first layer will be inclined. Therefore, adding a bracket at the bottom of the first layer provides excellent support. Do not worry about dumping and affecting the experience of blind users. As show in figure 30&31.

#### J. Testing

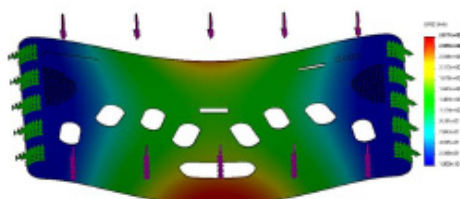


Figure 32. Solidworks deformation Analysis

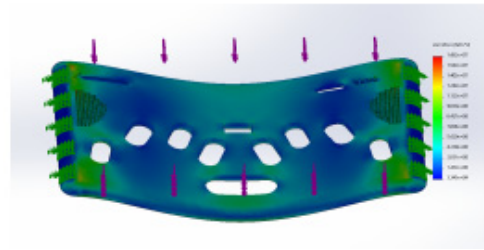


Figure 33. Solidworks stress Analysis

Detect the strength and service life of the product through top cover force analysis. Using ABS as a material can reduce the weight of the product. It was found that the part can withstand objects below 50N. ABS resin has good heat resistance, impact resistance, chemical resistance, low-temperature resistance and good electrical properties. As show in figure 32&33.

#### K. Virtual prototypes of the product





## V. CONCLUSION

The ageing of the population has become a big problem, and the number of blind people is increasing year by year in the context of the continuous development of intelligent communication. Therefore, providing an excellent digital reading platform for the blind can allow the disadvantaged to integrate into society.

This research comprehensively investigates touch reading products for blind people. Understand the development history of the blind reading device, the continuous development of braille and the advanced technology research of the braille reading device through literature review. Understand the daily life of the blind and experience the problems and difficulties encountered by the blind through observation. Understand big data and collect the basic preferences and usage habits of blind users, and design products to meet user needs through questionnaires. The interview method can truly experience the life of blind users. This method is very flexible, can understand things beyond the subject, and can understand the blind more truly. Case analysis analyzes the products on the market, mainly to summarize the functions of the products on the market, and find out their advantages and disadvantages. Finally, the design concept of this research is proposed.

The final reader design combines the trend of portable reading products, focusing on the design features of blind reading products. Through the use of vision and hearing, a variety of use modes are realized to meet the needs of blind people. Let blind people enjoy the same service resources, strengthen communication, and promote social construction.

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